

Operations of Functions

Let $f(x) = 2x^2 + 3x - 1$, $g(x) = x - 4$, and $m(x) = \frac{1}{x}$.

- 1) Find $(f + m)(0)$.
- 2) Find $(m - g)(2)$.
- 3) Find $(f + g)(-1)$.
- 4) Find $(f - m)(x)$ and state the domain.
- 5) Find $(f + g)(x)$.
- 6) Find $(f - g)(x)$.
- 7) Find $(f \cdot g)(x)$.
- 8) Find $\left(\frac{f}{g}\right)(x)$ and state the domain.

Let $f(x) = 2^x - 1$, $g(x) = 3\sqrt{x}$, and $h(x) = |x - 3|$.

- 9) Find $(f + g)(1)$.
- 10) Find $(h - g)(4)$.
- 11) Find $g(f(x))$.
- 12) Find $(f \circ h)(x)$.

Let $f(x) = x^2 + 1$, $g(x) = 3x - 2$, and $h(x) = \sqrt{x}$.

- 13) Find $(g \circ f)(2)$.
- 14) Find $(g \circ g)(x)$.
- 15) Find $(f \circ g \circ h)(4)$.
- 16) Find $(h \circ g \circ f)(x)$.

Use the tables below to answer questions 17 and 18.

- 17) Find $g(f(-3))$.
- 18) Find $\left(\frac{f}{g}\right)(4)$.
- 19) Find $h(f(3))$.
- 20) Find $(f \circ g)(4)$.
- 21) Find $(g \circ f \circ h)(3)$.
- 22) Find $f(f(-3))$.
- 23) Write $y = (3x - 1)^2$ as a composition of two or more functions.
- 24) Write $y = e^{5x+7}$ as a composition of two or more functions.
- 25) Write $y = \sin(x^2 + 4x)$ as a composition of two or more functions.
- 26) Write $y = \cos \sqrt{2x^3 + 1}$ as a composition of two or more functions.
- 27) If $f(x) = x^2 + 5x$ and $g(x) = 2x + 1$, find the values of x that make $f(x) = f(g(x))$ true.